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Model Question

Unit-1: Fundamentals of Data Communication and Layered Architecture

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Subject: Data Communication & Networks

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PART- A (2 MARKS)

1. A noiseless channel with a bandwidth of 6000 Hz is used for transmitting a signal with four signal levels. Calculate the maximum bit rate.
2. A channel has $B = 4$ KHz and a signal-to-noise ratio of 30 dB. Determine maximum information rate for 4-level encoding.
3. Write the key differences between half duplex and full duplex transmission modes with examples?
4. For “n” devices in a network, find the number of cable links required for a mesh, ring and star topology?
5. A signal travels from point A to point B. At point A, the signal power is 400 W. At point B, the power is 350 W. Determine the attenuation in decibels.
6. A noiseless channel with a bandwidth of 8000 Hz is used for transmitting a signal with eight signal levels. Calculate the maximum bit rate.
7. A signal travels through an amplifier, and its power is increased 10 times. In this case, calculate the amplification (gain of power) power?
8. For 8 devices in a network, find the number of cable links required for a mesh, ring and star topology?
9. A signal travels from point A to point B. At point A, the signal power is 200 W. At point B, the power is 170 W. Determine the attenuation in decibels.
10. Mention the advantages of optical fiber over twisted pair and coaxial cable.
11. The digital signal is to be designed to permit 160 kbps for a bandwidth of 20 KHz. Determine (a) number of levels and (b) S/N ratio.
12. A channel has $B = 10$ MHz. Determine the channel capacity for signal-to-noise ratio 60 dB.
13. What is crosstalk?
14. Let the energy strength at point 2 is $1/50$ th with respect to the point 1. Find out the attenuation in dB.
15. Assuming there is no noise in a medium of $B = 4$ KHz, determine channel capacity for the encoding level 4.
16. A channel has $B = 10$ MHz. Determine the channel capacity for signal-to-noise ratio 60 dB.
17. What is the difference between network layer delivery and transport layer delivery?
18. What are the responsibilities of the network layer in the Internet model?
19. What is the difference between a port address, a logical address, and a physical address?
20. If a periodic signal is decomposed into four sine waves with frequencies of 100, 300, 500, and 700 Hz, what is its bandwidth? Draw the spectrum, assuming all components have maximum amplitude of 15 V.

21. A non-periodic composite signal has a bandwidth of 400 kHz, with a middle frequency of 240 kHz and peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal.
22. The loss in a cable is usually defined in decibels per kilometer (dB/km). If the signal at the beginning of a cable with -0.35 dB/km has a power of 3 mW, what is the power of the signal at 6 km?
23. A telephone line has a bandwidth of 5000 Hz assigned for data communications. The signal-to-noise ratio is 3262. For this channel, calculate the capacity (C) of the channel?
24. How are OSI and ISO related to each other?
25. Name the four basic network topologies and cite an advantage of each type.
26. What does the Nyquist theorem have to do with communications?
27. What does the Shannon capacity have to do with communications?
28. What is quantization error? How can it be reduced?
29. Between RZ and NRZ encoding techniques, which requires higher bandwidth and why?
30. What is the advantage of delta modulation over PCM?
31. Distinguish between a signal element and a data element.
32. Compare and contrast PCM and DM.
33. Write down the different steps of Pulse code modulation (PCM) scheme to convert the analog signal to its digital form?
34. Illustrate about different Transmission modes.

PART- B (7 MARKS)

1. Discuss the seven-layer functionalities of the OSI reference model with neat diagram.
2. Encode the given data word 100110101 using NRZ-L, RZ, Manchester, Differential Manchester,
3. What are the layers of the OSI model? Write down the responsibility or key concern of each layer of the OSI model? What is the difference between TCP/IP and OSI model?
4. Encode the given data word "10101001" using Unipolar NRZ, NRZ-L, NRZ-I, RZ, Manchester, Differential Manchester coding, and Bipolar AMI techniques.
5. Encode the given data word "0110101001" using NRZ-L, NRZ-I, RZ, Manchester, Differential Manchester, Bipolar AMI and MLT-3 coding techniques.
6. Explain the different stages of Pulse Code Modulation (PCM) technique with neat diagram to convert the analog signal to digital signal.
7. Name the four basic network topologies, and cite an advantage of each type. We have two computers connected by an Ethernet hub at home. Is this a LAN or WAN? Explain the reason.
8. What is TCP/IP protocol suite? With a diagram shows the logical connection between layers of a TCP/IP protocol suite. Discussed the different layers of TCP/IP model along with protocols?
9. Illustrate in details about the different types of transmission impairment.